MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TYPE QK427

The complete requirements for procuring the electron tube described herein shall consist of this document and the issue in effect of Specification MIL-E-1.

Description: Magnetron, 10 Watts (minimum), fixed frequency 10,050 to 10,250 megacycles, CW air-cooled, integral magnet, unipotential cathode.

Dependent Ratings: (Note 22)	Ib mAdc	Pi W
Absolute Maximum:	150	210
Minimum:	---	---

Independent Ratings (Note 23)	Ef V	Eb kVdc	tk sec	O'	Anode T. °C.	Output pressurization	Environment pressurization
Absolute Maximum:	5.5	1.4	---	1.5	120	45 psia	45 psia
Minimum:	--- Note 2 Note 11	---	180	--- Note 3	--- Note 4	--- Note 5	700mm Hg(absolute) Notes and 6

Input bushing: Note 7
Cooling air: Note 7
Magnet isolation: Notes 7 and 8
Mounting support: Note 7
Mounting position: Cathode vertical
Output coupling: Note 7
Input connections: Note 7
Vibration, shock: Note 9
Weight: 3-5/8 pounds, approx.

Ref.	Test	Conditions	Symbol	Limits Min.	Limits Max.	Unit
	General					
3.1	Qualification approval	Required; Note 21				
4.9.2	Dimensions	Per Drawing RE38D538				
3.7	Marking	Note 1				
4.5	Holding period	t = 168 hours				
4.9.18.1.8	Container drop	Required				
	Qualification inspection (Note 25)					
4.9.8	Salt spray corrosion	Omit				

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Ref.	Test	Conditions	Symbol	Limits Min.	Limits Max.	Unit
	Qualification inspection (Note 25) (Cont'd)					
4.9.19.2	High-frequency vibration	No voltage				
---	Shock test	Note 10; G = 15				
	Acceptance inspection, part 1 (production)					
4.10.8	Heater current	Ef = 4.5 V Note 11, tk = 300		3.5	4.3	A
4.16.3	Oscillation 1	Note 13				
---	Standing wave ratio	σ' = 1.1 max.				
4.16.3.2	Heater-cathode warm-up time	tk = 180 sec. max. at Ef = 5.0 V, Ef for test 4.5 volts				
4.16.3.4	Average anode current	Ib = 115 mAdc				
4.16.3.6	Power output	Method B	Po	13	---	W
---	Anode voltage		Eb	1.05	1.40	kVdc
4.10.7.3.1	Fixed tuned frequency		F	10,050	10,250	Mc
4.16.7.2	Load stability	Note 15				
---	Amplitude noise	Notes 12, 16, 17 and 18	AM Noise	---	15	db
---	Coherent amplitude noise	Note 19	Amplitude level	---	6	db
	Acceptance inspection, part 2 (design)					
4.9.19.1	Low-frequency vibration	No voltage				
4.16.5	Pulling factor	σ' = 1.5, Note 14	ΔF	---	20	Mc
	Acceptance inspection part 3 (life)					
4.11	Life tests	Group D	T	200	---	hours
4.11.4	Life test end point	Notes 16, 17 18 and 20	F Po Eb A. M. Noise	10,000 10 1.0 ---	--- --- 17	Mc W kVdc db
5.1	Preparation for delivery	Note 24				

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Note 1: The tube shall be marked "USN-QK427".

Note 2: The maximum value specified herein is for a nonoscillating condition. Heater surge current shall not exceed 7.0 amperes.

Note 3: Frequency skipping or unstable operation may be encountered at some phase positions when the mismatch occurs at the end of a "long" line.

Note 4: The temperature shall be measured at the point shown on Drawing RE38D538.

Note 5: The gas used shall provide insulating properties at least equal to that of clean, dry air at the pressures specified herein.

Note 6: The tube shall be fitted with the connector conductor assembly, or its equivalent, as shown on Drawing RE38D538.

Note 7: Drawing RE38D538 shall apply.

Note 8: In handling and mounting the magnetron, care shall be taken to prevent demagnetization. Ferromagnetic materials or energized magnets shall not be brought within 2 inches of the tube.

Note 9: Care shall be used in the storage, installation, and handling of the tube to avoid imparting vibration or shock in excess of the values which it is designed to withstand.

Note 10: The magnetron shall be mounted on a test plate and dropped five times on each of three mutually perpendicular axes parallel to the reference planes shown on Drawing RE38D538. The shock pulse shall have a duration of approximately 11 milliseconds as measured at the quarter amplitude points of the acceleration shock wave.

Note 11: Standby time with the heater voltage at the value specified herein for warmup shall be kept at a minimum.

Note 12: The average anode current and heater voltage specifying the center of the quiet area shall be marked on each tube.

Note 13: The tube shall be operated from a current regulated power supply having a static impedance greater than 100,000 ohms and a dynamic impedance equal to the reactance of a 4 μf condenser.

Note 14: The pulling measurement shall be made in such a manner that thermal effects do not introduce appreciable errors.

Note 15: The magnetron shall operate satisfactorily without shifting modes while a 1.5 to 1 VSWR mismatch is shifted through all phase positions.

Note 16: The quiet area of operation is defined as a region wherein the limits of noise measured as specified in Note 17 are not exceeded. This region shall exist over a 10 mAdc minimum range in anode current between 85 and 145 mAdc and over a 0.5-volt minimum range in heater voltage between 0 and 5.0 volts. The quiet area shall also exist when a load of minimum VSWR of 1.2 to 1 is varied through all phase positions.

Note 17: (a) In conducting this test, a receiving system consisting of a crystal detector, a video amplifier, and a Vacuum Tube Volt Meter (VTVM) shall be used. The video amplifier shall have the following characteristics:

(1) Input impedance: approximately 300 ohms.
(2) Gain: variable to plus 100 db.
(3) Response: 2 ± 1 to 50 ± 10 kc, at 3 db points.

(b) The VTVM shall be connected at the video amplifier output. The measurement shall be conducted as follows:

(1) With no voltage applied to the magnetron, the video amplifier gain shall be adjusted for a VTVM reading of 0.1 volt r.m.s. This level shall be used as a zero db reference point.

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(2) With the magnetron oscillating, r.f. energy shall be coupled through a 1N23C crystal detector to the video amplifier input, the r.f. input being adjusted to obtain 0.5 mAdc of crystal current.
(3) The increase in signal level indicated by the VTVM shall be taken as the measurement of AM noise.

Note 18: This measurement includes noise generated by the crystal detector, and therefore shall be minimized by the careful selection of quiet crystals. A quiet crystal is defined as one which has a maximum of 10 db noise rise when measured as follows:

(a) The crystal under test shall be connected to the input of a video amplifier as specified in Note 17.
(b) A VTVM shall be connected to the output of the video amplifier.
(c) With no r.f. or d.c. energy applied to the crystal, the video amplifier gain shall be adjusted for a VTVM reading of 0.1 volt r.m.s. This level shall be used as a zero db reference point.
(d) With a d.c. voltage applied to the crystal for a current of 0.5 mAdc, the noise rise measured shall not exceed 10 db.

Note 19: The video amplifier specified in Note 17 shall be scanned with a 1 kc filter from approximately 1 to 20 kc at an approximate rate of 7 c.p.s. and displayed on an oscillograph. No coherent signal shall be apparent which has a duration of more than 1 second, and an amplitude greater than 6 db above normal noise as observed on the oscilloscope. Operation shall be observed over any 2-minute interval within a 6-minute period. The requirements of this specification sheet shall be met over the minimum quiet area specified in Note 17.

Note 20: To maintain the noise factor throughout life, it may be necessary to adjust both the heater voltage and the plate current from the value specified in Note 12 to new values within the limits specified in Note 15 due to the shifting of the quiet area.

Note 21: The activity responsible for the qualified products list is the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification". (Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pa.)

Note 22: Caution. - The dependent absolute ratings are interrelated, and it does not necessarily follow that combinations of ratings can be attained simultaneously. The provisions of paragraph 6.5 of Specification MIL-E-1 shall apply in the selection of the operating point.

Note 23: The independent absolute ratings shall not be exceeded if the life specified herein is to be obtained. These independent absolute ratings are limiting values beyond which the serviceability of any individual tube may be impaired.

Note 24: Tubes shall be packaged and packed as specified in the contract or order in accordance with Specification MIL-E-75/9. Rough handling test (i) shall apply.

Note 25: These tests are performed only during qualification inspections; however, qualification inspection shall consist of all tests listed on this specification sheet and the applicable tests of Specification MIL-E-1.

Note 26: This specification supersedes Specification MIL-E-1/789A(NAVY), 27 April 1958. Specification MIL-E-1/789A(NAVY) and all previous issues have been declassified.

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